

EC466: Estimation and Inference in Econometrics

Exercise # 1

You need to install Python on your computer to solve Question 1. To install Python please go to <https://software.essex.ac.uk/> and search for Python or Anaconda. Anaconda is a larger distribution package.

1. Please answer the first 15 questions in EC466hw1.py, which is a python script file. If you are not familiar with Python then there are many Youtube resources that are useful. It's worthwhile knowing the fundamentals of the language before relying on AI tools. Please copy and paste **attend.xls** file in a suitable root folder; otherwise, you will receive an **error message**. For instance, I saved **attend.xls** file in the following folder “/Users/etheridge/Documents/”.
2. Show that $\mathbf{X}'\mathbf{X}$ is a symmetric positive definite matrix.
3. Show that $(\mathbf{X}'\mathbf{X})^{-1}$ is a symmetric positive definite matrix.